

16.2 Troubleshoot IPv4 Static and Default Route Configuration

Troubleshoot IPv4 Static and Default Route Configuration Network Changes

Networks fail for a number of reasons:

- An interface can fail
- A service provider drops a connection
- Links can become oversaturated
- An administrator may enter a wrong configuration.

Network administrators are responsible for pinpointing and solving the problem.

To efficiently find and solve these issues, it is advantageous to be intimately familiar with tools to help isolate routing problems quickly.



Troubleshoot IPv4 Static and Default Route Configuration

Common Troubleshooting Commands

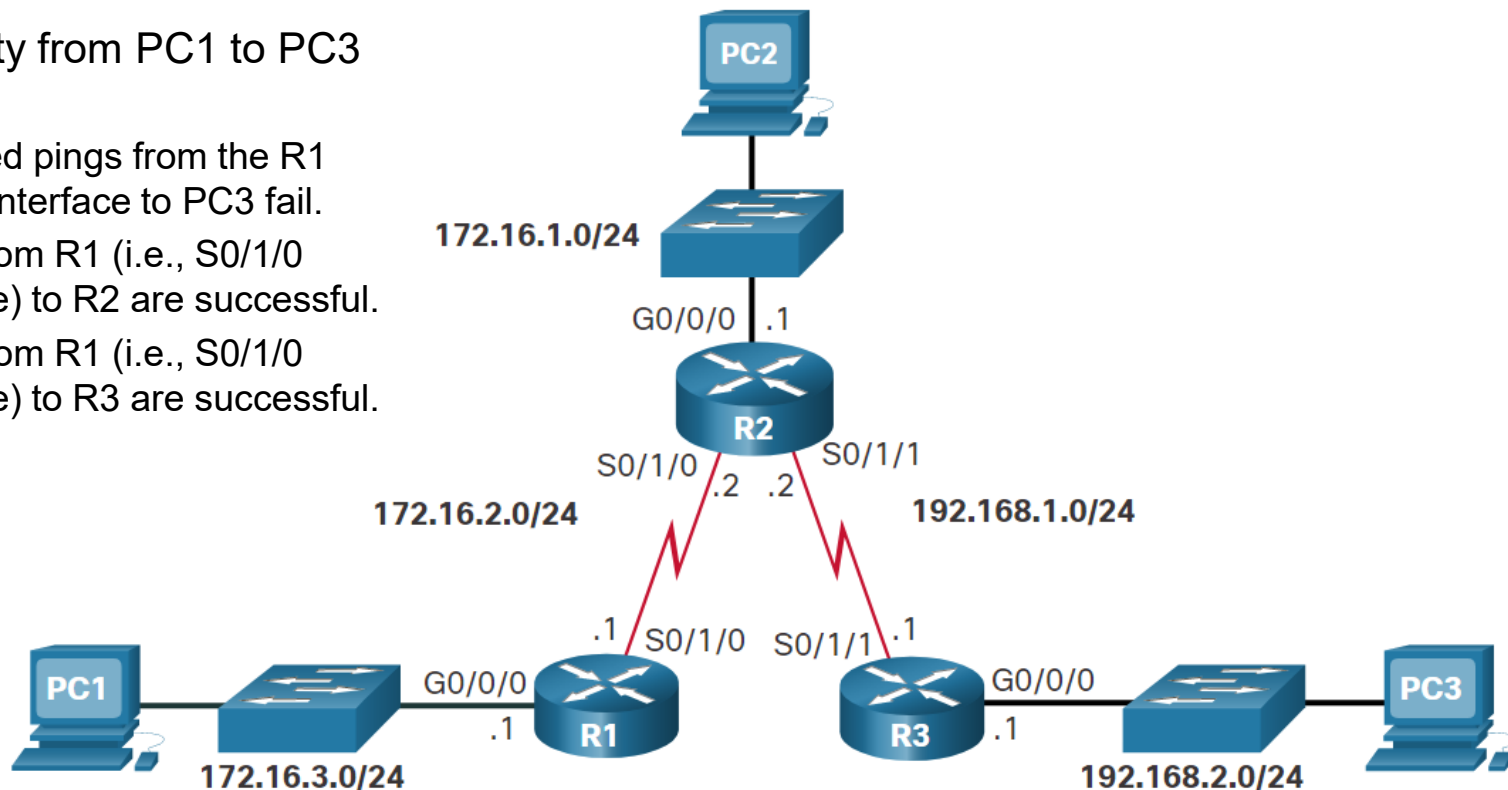
Command	Description
<code>ping</code>	• Verify Layer 3 connectivity to destination. • Extended pings provide additional options.
<code>tracert</code>	• Verify path to destination network. • It uses ICMP echo reply messages to determine the hops to the destination.
<code>show ip route</code>	• Displays the routing table. • Used to verify route entries for destination IP addresses.
<code>show ip interface brief</code>	• Displays the status of device interfaces. • Used to verify the operational status and IP address of an interface.
<code>show cdp neighbors</code>	• Displays a list of directly connected Cisco devices. • Also used to validate Layer 1 and 2 connectivity.

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Solve a Connectivity Problem

Connectivity from PC1 to PC3 fails.

- Extended pings from the R1 G0/0/0 interface to PC3 fail.
- Pings from R1 (i.e., S0/1/0 interface) to R2 are successful.
- Pings from R1 (i.e., S0/1/0 interface) to R3 are successful.

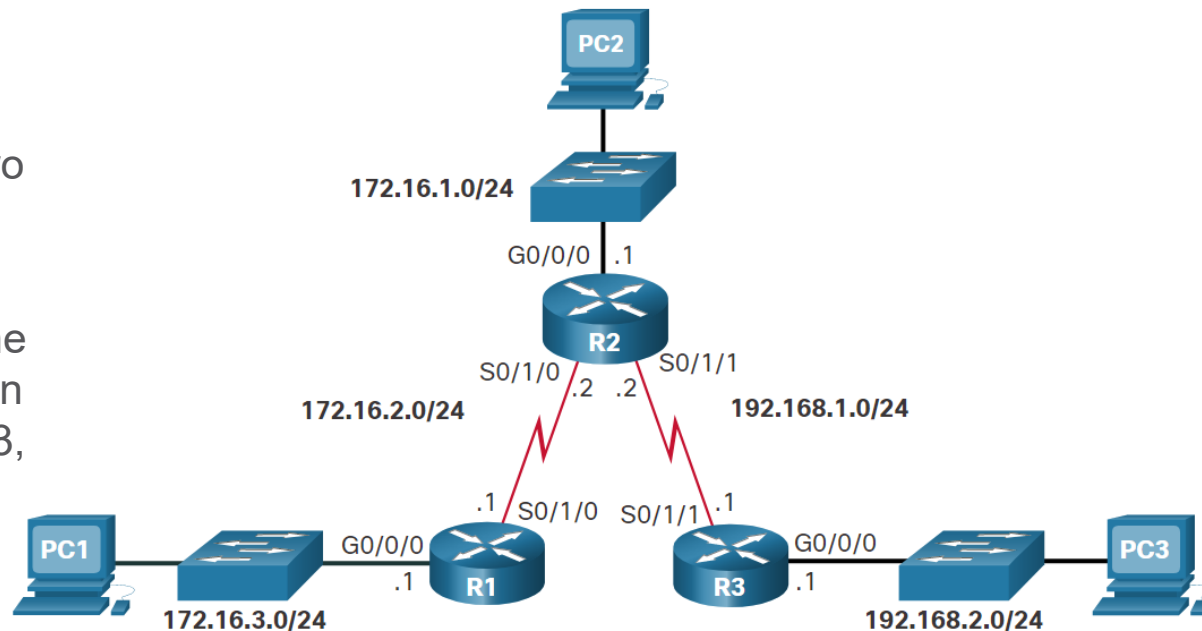


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Ping The Remote LAN

The network administrator can test connectivity between the two LANs from R1 instead of PC1.

This can be done by sourcing the ping from the G0/0/0 interface on R1 to the G0/0/0 interface on R3,



```
R1# ping 192.168.2.1 source g0/0/0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.3.1
.....
Success rate is 0 percent (0/5)
```

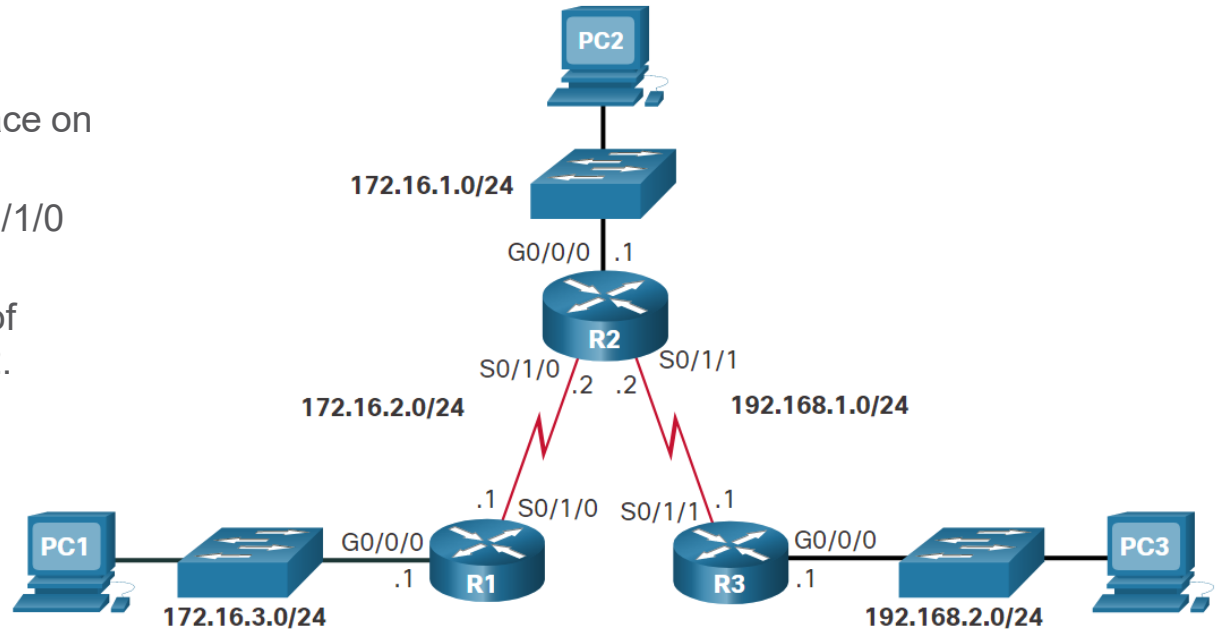
Troubleshoot IPv4 Static and Default Route Configuration

Ping the Next-Hop Router

Next, a ping to the S0/1/0 interface on R2 is successful.

This ping is sourced from the S0/1/0 interface of R1.

Therefore, the issue is not loss of connectivity between R1 and R2.



```
R1# ping 172.16.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/3/4 ms
```

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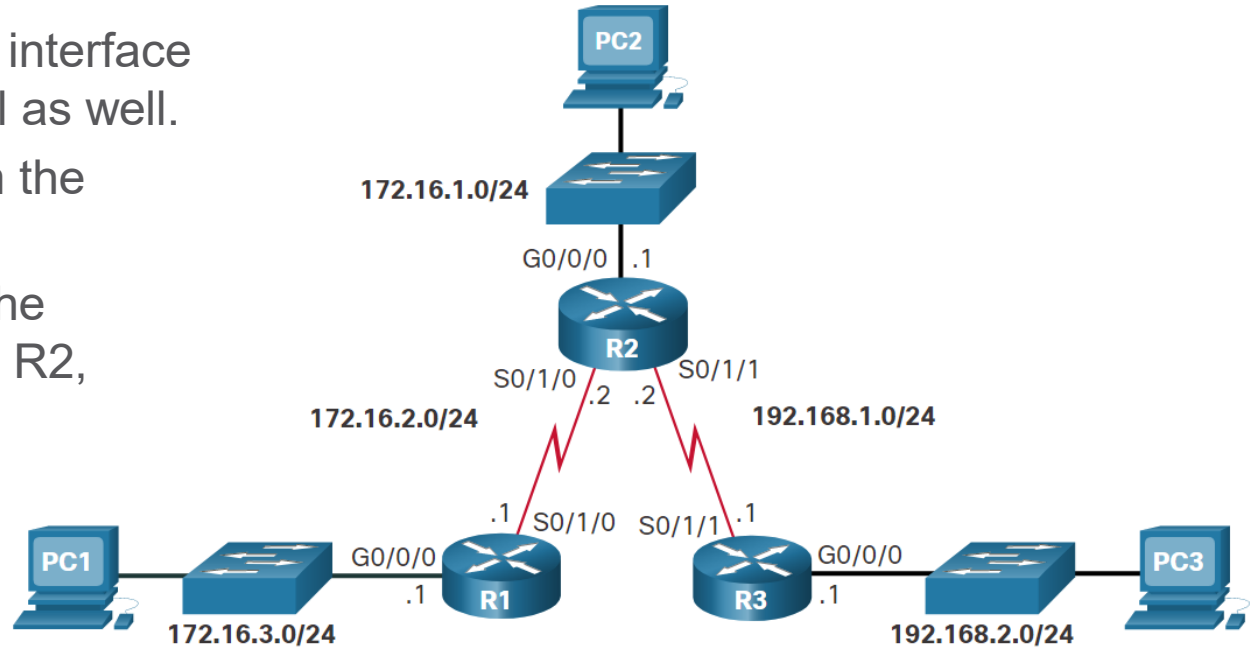
Ping R3 LAN from S0/1/0

A ping from R1 to the R3 interface 192.168.2.1 is successful as well.

This ping is sourced from the S0/1/0 interface on R1.

R3 has a route back to the network between R1 and R2, 172.16.2.0/24.

This indicates that either R2 or R3 may have an incorrect or missing route to the LAN on R1.



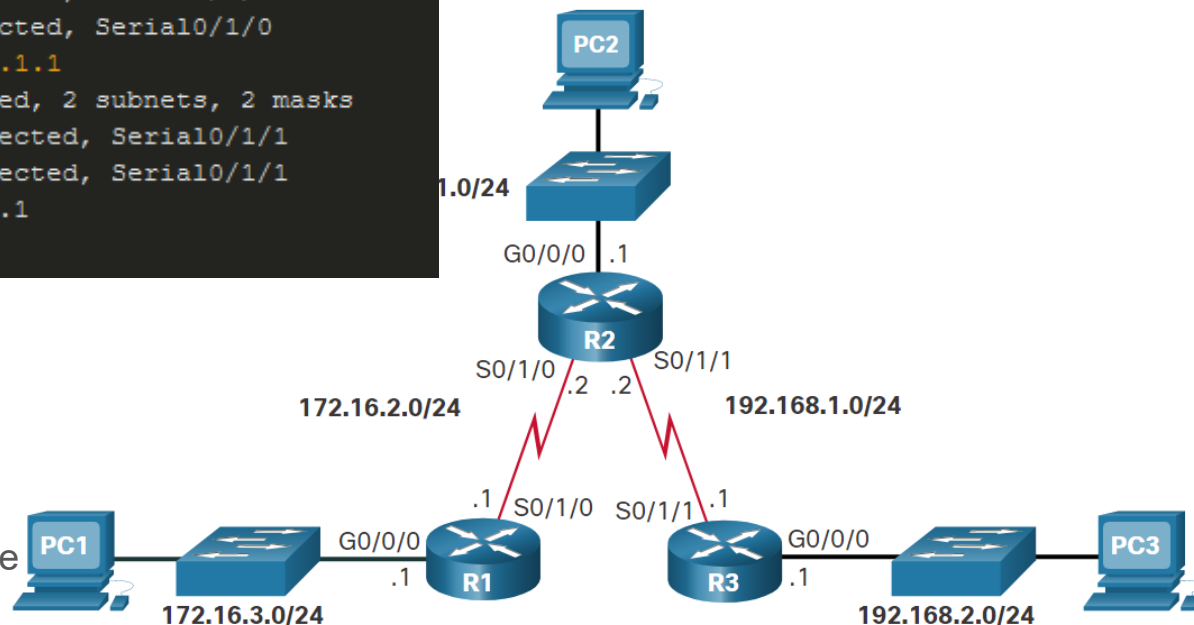
```
R1# ping 192.168.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/3/4 ms
```

Troubleshoot IPv4 Static and Default Route Configuration

Verify the R2 Routing Table

```
R2# show ip route | begin Gateway
Gateway of last resort is not set
  172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
C       172.16.1.0/24 is directly connected, GigabitEthernet0/0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0/0
C       172.16.2.0/24 is directly connected, Serial0/1/0
L       172.16.2.2/32 is directly connected, Serial0/1/0
S       172.16.3.0/24 [1/0] via 192.168.1.1
  192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, Serial0/1/1
L       192.168.1.2/32 is directly connected, Serial0/1/1
S       192.168.2.0/24 [1/0] via 192.168.1.1
R2#
```

- The next step is to investigate the routing tables of R2 and R3
- Notice that the 172.16.3.0/24 network is configured incorrectly.
- The static route to the 172.16.3.0/24 network has been configured using the next-hop address 192.168.1.1.



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Correct the R2 Static Route Configuration

Next, the running configuration does, in fact, reveal the incorrect **ip route** statement

The incorrect route is removed, and the correct route is then entered.

```
ip route 172.16.3.0 255.255.255.0 172.16.2.1
```

```
R2# show running-config | include ip route
ip route 172.16.3.0 255.255.255.0 192.168.1.1
ip route 192.168.2.0 255.255.255.0 192.168.1.1
R2#
R2# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)# no ip route 172.16.3.0 255.255.255.0 192.168.1.1
R2(config)# ip route 172.16.3.0 255.255.255.0 172.16.2.1
R2(config)#
```

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Verify New Static Route is Installed

The routing table on R2 is checked once again to confirm the route entry to the LAN on R1, 172.16.3.0, is correct and pointing toward R1

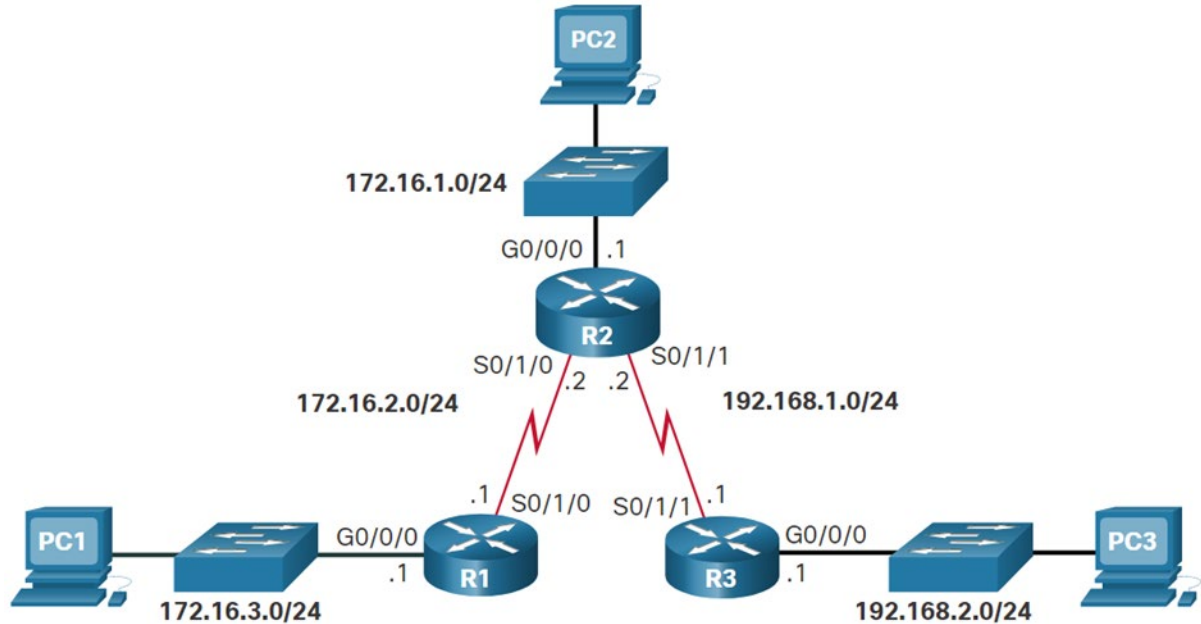
```
R2(config) # exit
R2#
*Sep 20 02:21:51.812: %SYS-5-CONFIG_I: Configured from console by console
R2# show ip route | begin Gateway
Gateway of last resort is not set
    172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
C       172.16.1.0/24 is directly connected, GigabitEthernet0/0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0/0
C       172.16.2.0/24 is directly connected, Serial0/1/0
L       172.16.2.2/32 is directly connected, Serial0/1/0
S       172.16.3.0/24 [1/0] via 172.16.2.1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, Serial0/1/1
L       192.168.1.2/32 is directly connected, Serial0/1/1
S       192.168.2.0/24 [1/0] via 192.168.1.1
R2#
```

Troubleshoot IPv4 Static and Default Route Configuration

Ping the Remote LAN Again

Next, a ping from R1 sourced from G0/0/0 is used to verify that R1 can now reach the LAN interface of R3.

As a last step in confirmation, the user on PC1 should also test connectivity to the 192.168.2.0/24 LAN.



```
R1# ping 192.168.2.1 source g0/0/0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.3.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/4/4 ms
```